

Radar Search Pipeline TO BE 0.7.6.3.6.12

Review design for a planned Radar search pipeline change

Status: TO BE

Slice: 0.7.6.3.6.12: Review-needed entity projection and evaluation matcher parity

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1. Decision Context

The last Docker benchmark_smoke_plus proved that recall expansion can now find the remaining Tobolsk production-site evidence. The run still reported tobolsk-site as a false negative because the found review-needed entity was not evaluated as a production-site match.

This is no longer a scheduler, selection, or provider-search problem. It is a handoff problem between upstream disambiguation, candidate-universe projection, and offline evaluation.

2. AS IS Problem

Current behavior:

1. Expansion retrieves source-backed material about Tobolsk.
2. Upstream disambiguation can represent Тобольская промышленная площадка as entity_type=production_site.
3. A candidate-universe gap with the same name can enter the universe first without entity type metadata.
4. Candidate-universe metadata enrichment defaults missing metadata to unknown_entity.
5. The later typed upstream entity is skipped as duplicate because the name is already present.
6. Evaluation sees either unknown_entity or a name variant that is too strict to match the curated baseline.

The result is misleading: Radar did find the object, but evaluation reports it as missed.

3. Intended Pipeline Behavior

Review-needed upstream entities must preserve their semantic type across the pipeline:

- production_site, branch, asset, and project must not degrade to unknown_entity when the type is already known.
- If an untyped universe row already exists and a typed upstream row arrives later, the existing row must be upgraded in place.
- resolution_status, resolved_legal_name, not_candidate_reason,

review_flags, and source_refs must be merged rather than dropped.

- Product candidates remain strict and are not changed by this slice.
- Evaluation counts review-needed production-site matches through

candidate_universe, upstream_disambiguation_results, or linked facts when the name/alias match is source-backed.

4. Roles Changed

Role	Change
Candidate universe support	Preserve entity type and review metadata for review-needed upstream entities.
Staged execution projection	Merge typed upstream metadata into existing universe rows instead of skipping duplicates.
Evaluation	Match production-site baseline entities against review-needed universe/upstream entities with tolerant source-backed name matching.
Dossier/report	Continue exposing product candidates strictly, while review recall can use diagnostic universe entities.

5. Context Passed Between Roles

The following fields must survive the projection boundary:

- entity_type
- resolution_status
- resolved_legal_name
- not_candidate_reason
- review_flags
- source_refs
- linked_legal_name when available

If the same entity appears in several diagnostic lists, the more specific metadata wins over unknown_entity and empty fields.

6. Flow Diagram

7. Evaluation Semantics

Evaluation should remain offline and provider-free.

For non-legal baseline entities:

- exact normalized name remains strongest;
- alias match remains valid;
- source-backed partial match may be medium-confidence;
- production-site wording can tolerate suffixes such as (Дирекция компании)

or extra relation words such as СИБУР.

This is not SIBUR hardcoded. It is generic review-needed site matching over the curated baseline aliases and source-backed observed names.

8. Diagnostics

False-negative diagnostics should distinguish:

- `not_retrieved_in_run`: no source text or observed entity exists;
- `present_not_projected`: source text exists, but no observed entity exists;
- `projection_type_lost`: observed entity exists but type was downgraded to `unknown_entity`;
- `present_not_matched`: observed entity exists with a type/name mismatch;
- budget/selection buckets for targets that were never executed.

9. Test Plan

Unit tests:

- candidate-universe metadata enrichment preserves an existing `production_site` type when observation metadata is absent;
- a typed upstream entity upgrades an existing duplicate `unknown_entity` row;
- gap payloads preserve `entity_type`, `not_candidate_reason`, and `review_flags`;
- evaluation matches `Тобольская промышленная площадка (Дирекция компании)` to `Тобольская промышленная площадка СИБУР`;
- evaluation reports `projection_type_lost` when only an `unknown_entity` observed row exists for a production-site baseline.

Regression tests:

- `tests/test_radar_evaluation.py`
- `tests/test_live_icp_radar.py`
- `tests/test_radar_benchmark.py`
- architecture and documentation contracts.

10. Acceptance Criteria

- If a Docker smoke run finds source-backed Tobolsk production-site evidence, `tobolsk-site` is not reported as false negative due to projection loss.
- `review_recall` reaches 1.0 when all three production-site baseline entities are present as review-needed or linked upstream entities.
- `Product /candidates` remains strict.
- `benchmark_live` remains blocked until bounded smoke is fully interpretable.

11. Out Of Scope

- No scoring changes.
- No scheduler or budget changes.
- No provider changes.
- No UI changes.
- No SIBUR-specific production runtime hardcode.